Masters Thesis proposal

Žan Žagar

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1 Introduction

1.1 Context and scope

sonar imaging technology is applied by the Dutch Police in various underwater search scenarios. Traditionally sonar operators are employed to detect the target objects from sonar readings. Finding drowning victims is one of the tasks of the Landelijk Team onderwaterzoekingen and due to the constant attention required by the sonar operator going trough vast stretches of canals it is very labour intensive. Additional problems arise due to the physical properties of the sonar acoustic signal, and how they interact with a humany body. Because a human body contains a large amount of water, they are often better found either by their acoustic shadow or the objects with better acoustic reflective properties around them. The shadow from a recent victim will generally be small and increase only if decomposition and subsequent bloating of the decomposed body affect the size and raise it above the bottom [4]. Skeletonised bodies can be identified in the same way but the smaller size of the bones requires much higher resolutions than traditionally used. The task of human body detection using sonar is therefore difficult especially in complex terrains [4].

Computer vision tasks seek to enable computer systems automatically to see, identify and understand the visual world, simulating the same understanding as human vision does. [7]. Object detection presents one of the most fundamental and challenging problems in computer vision, it seeks to locate object instances from a large number of predefined categories in natural images [40] [23].

In recent years the computer vision field has benefited greatly from deep learning advances, in the field of underwater environments modeled using acoustic means (sonar) approaches using deep learning show very promising results [9] [17].

Deep learning approaches are not well established for object detection on acoustic images such as sonar data, mainly due to the problem of lacking training data. However recently approaches using convolutional neural networks and simulated data have been gaining more and more success with this task [9]. Many factors such as speed, limited data, and class imbalances make object detection in general a very challenging task. In the domain of underwater object detection this problem is compounded by factors such as speckle noise, reverberation, acoustic shadowing and limitations of the readings being 2D horizontal projections of the environment (Figure 1) [17]. Additional problems arise due to the dual priority nature of the task, because we need to not only localise the object, but also classify it [17].

For detecting underwater objects the limited resolution and very high noise levels in the underwater environment are the main problem and this is especially the case when using side scan sonar, which is prone to noise. This gets compounded by the fact that we are trying to detect objects that are not rigid bodies (drowning victims) [33].

Concerning the problems of limited training data and class imbalances, these have been tackled with some success using ray tracing simulators to simulate

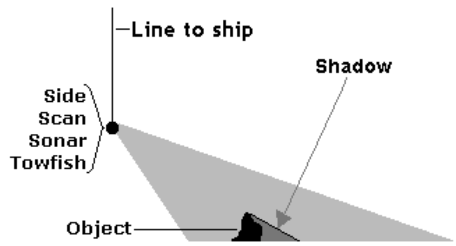

Figure 1: How side scan sonar projects the 2D image, objects in the shadow of another object can not be detected (source: https://dosits.org/people-and-sound/navigation/how-is-sound-used-to-find-objects-on-the-ocean-bottom/)

the properties of acoustic sound waves hitting an object and casting a sonar shadow and then using GANs and style transfer neural networks to generate simulated data that is consistent with the training data [31]. Other approaches only used a style transfer network trained on a set of styles and then applied these to a manipulated gray-scale image where a sonar shadow mask is added to it to produce simulated sonar images [9] [20]. Along with transfer learning, which allows the model to learn low level features on a bigger dataset, then be fine-tuned for the detection task these approached have produced positive results for object detection in underwater scenarios [9].

1.2 Research scope

Our assigned task tackles the problem of creating a sonar reading based object detector for detecting drowning victims in the shallow water canal environment. We want to analyze and see if our methods and different environment compared to other succesful approaches [13] [9] [20] can still produce adequate results compared to the state of the art.

1.2.1 Shallow waters and varied beddings

Since related work ([13] [9] [20]) focused on sonar readings in deep sea environments, it is not clear whether these approaches would also be succesful in an environment with a different acoustic profile. THe most important questions are what effect on learning performance these factors will have. For example since the waters might be shallow, one could assume that since the returned sonar wave took a shorter total path, the signal will be stronger, and therefore the readings might be less noisy. However this does not take the reflective properties of materials in canals into account. Muddy and rocky water floors reflect sound differently, and this has a big impact on the noise level in sonar readings.

1.2.2 Available side scan data

The dataset has been provided by the Dutch police, it includes readings by the 4125i side scan CHIRP sonar probe. Side scan sonar is very sensitive and can therefore detect very small complex objects. That is why it is typically used for tasks involving detecting objects on the water floor [33]. The aforementioned sonar probe was towed behind a police boat, which produced 300 gigabytes of sonar readings which are in the EdgeTech .jsf format.

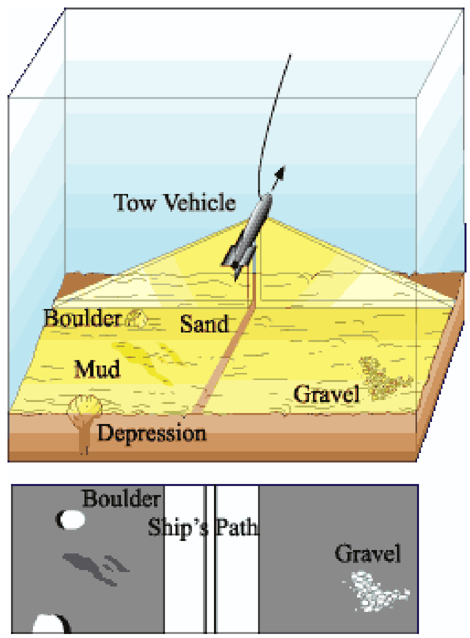

Figure 2: The top image shows the underwater scene in 3D, the bottom shows what the side scan sonar projection looks like. Diagram courtesy of USGS Seafloor Mapping Technology group, woodshole.er.usgs.gov/operations/sfmapping

1.3 Research question

In this work we propose an approach to detecting drowning victims using deep learning based on sonar images and data simulation.

1. Does using a deeper backbone (resnet50) with the pre-trained Faster R-CNN model increase overall accuracy and recall for drowning victim detection when compared to the same model using a shallower (resnet19) backbone? Deeper networks have better feature extraction capabilities but are prone to overfitting, we want to see how beneficial they are compared to a shallower network which is less prone to overfitting but has lesser feature extraction capabilities, therefore we will train the two mentioned models using only real data

and compare their overall accuracy and recall on the validation set.

While neural style transfer has been successful for generating training data for sonar images [20],[9] the problem of choosing style images that sufficiently represent the environment remains.

2. Can using a cycleGAN to generate synthetic data improve our accuracy and recall compared to using fast neural style transfer with a pre-determined set of style images? We will answer this question by training two variations of the best performing model of question 1. each on synthetic data to real data ratios of 1 in 5 (synthetic) one produced by the fast NST algorithm, the other by the cycleGAN. We will compare them using overall accuracy and recall.

3. What kind of ratios of synthetic to real data give the biggest learning performance benefit? This will be evaluated by measuring what effect on overall accuracy and recall using 1 in 2, 1 in 5 and 1 in 10 ratios of synthetic to real data have compared to each other, again we will use the best performing model of question 1 to answer this.

4. How big is the performance loss when training our model only on synthetic data? For this we want to train our best performing model again but this time only with synthetic data, and see what kind of performance (measured with overall accuracy and recall) is achieved when compared to our best performing baseline model, and our best performing synthetic ratio model.

5. What effect does the muddy shallow water environment have when compared to successful sea side scan sonar approaches? Given the different environment and noise characteristics of the Dutch canals, we want to compare our results to some of the state of the art work produced in similar approaches [20] [9] [31] [13], and see how our various performance measures (F1 score, precision, recall accuracy) compare to theirs when our model is trained using their data or when their model is trained using our data.

1.4 Structure

This work has the following main chapters: literature, method, experiment and conclusions. The literature chapter lays out the required background knowledge for research in this field, the problem and the related work that tried tackling this problem. The method chapter will describe our task from the police, our chosen approach of tackling the problems associated with this task, and what related works motivated us in choosing our selected approach. Our experiment chapter will analyze the results we obtained and compare various models we used between each other, and also use various metrics to analyze their performance compared to the state of the art. Finally the conclusion chapter will include a discussion of our obtained results, our findings, and how we interpret them in relation to other work that was done in the field.

2 Literature review

In the following chapter we will discuss various aspects of sonar first, such as its applications, its physical properties, the types of sonar that is used and their features. We will then describe automated detection from sonar images as it relates to our problem, the architectures of object detection models, the R-CNN family of models and considerations when choosing a classification head for an object detection model. That will be followed up by an introduction to synthetic data generation and an overview of what has recently been done in that area and has proven successful in the field of deep learning applied to underwater acoustic data.

2.1 Sonar

sonar (sound operated navigation and ranging) is a technique that uses sound (echolocation) for detecting and ranging objects. Submarines submerged beneath the ice at the north pole for example cannot use light to navigate, so they use very high powered sonar pings (in the form of 235-decibel pressure waves) to range, detect objects and navigate what their environment looks like. The sound hitting an object also casts a "sonar shadow" behind the object, from which we can predict various features of the target object itself (size, angle etc) [6] [5] [34].

Due to its versatile nature and the physical properties of sound it is frequently used to localize and find objects that are underwater (searching for ship, plane wrecks or drowning victims) and estimate the depth of large bodies of water [9].

2.1.1 Sonar types and characteristics

There are two types of sonar, active sonar and passive sonar. Active sonar is in principle similar to the animal echo location example given above, where a sound wave is sent out, and we wait for the response to measure the distance of the reflected object. Passive sonar is essentially just listening for sound without emitting sound. Passive sonar can be used for example in the military for submarines to avoid detection, but its main drawback is that estimating object positions becomes much harder as you can not as easily compute how long the sound has traveled because you are not the source. For the application of this work our discussion will mainly be focused on active sonar. Active sonar works by sending out a pulse from a transmitter and receiving it by a receiver, which allows for more accurate object distance estimation. [18] [6] [34]

While sonar is very useful for low or no light environments, it also has its drawbacks. Namely with active sonar there is the ethical issue of danger to wildlife, where powerful sound waves can disturb or kill marine wildlife. There are also problems that relate to noise given an environment of many objects that can also reflect sound and cause reverberation. Schools of fish could block detection of an object under them due to their sonar shadow for example, muddy seafloors or porous objects may not have good sound reflecting properties [33].

2.1.2 Physical sonar properties

Frequency is the number of times in a second that an emitted sound wave repeats itself. Lower frequency waves can cover a larger area, but will not penetrate very deeply (therefore they will produce readings with more noise). Higher frequency waves will only cover a narrower area but penetrate deeply (more detail), this is because of the physical properties of how water molecules absorb or allow sound to pass trough them. It is explained below in the context of sonar sound absorption [34]. There is also a delay between sonar pings, where the system is waiting for the echo. To get over both of these limitations a technique called CHIRP (Compressed High-Intensity Radiated Pulse) emerged in the 1990's with the objective of remote acoustic classification of seafloor sediments. CHIRP sends out pulses of varying frequency ranges that are usually around 10 times the duration of a traditional fixed frequency sonar ping. This allows for much more accurate and more detailed readings of the modeled environment because instead of the ping it sends out a longer pulse of varying frequency. [19] Using CHIRP allows for clearer high resolution readings, however a few limitations still remain. One is that in sonar every sound that is returned from another source other than the target of interest is referred to as noise or reflection (reverberation). These can include fish, marine life, small particles and inconsistencies in the water etc. Reverberation is especially pronounced in shallow waters (like canals) [3]. Vessel and water movement also cause noise in the sensor readings, due to deviations in the sound origin.

Objects from which the sound reverberates also produce a sonar shadow area behind them. Sonar shadow is the area behind an object that is hit by the sound wave, which the sound wave cannot reach as it is reflected back from the object [34]. When visualized this looks like a shadow outline of the object, which in the case of the object being our detection target, helps us to discern its shape and angle. This can be seen from the image in Figure 3

Another problem related to noise mainly in deeper waters is that of refraction, since there is often a change in temperature and pressure, with the change of depth (high pressure, faster sound speed) colder still waters cause sonar to refract when it is emitted from the warmer moving surface temperature waters [34].

2.1.3 Sonar propagation properties

The performance of sonar is very dependent on environmental factors such as sound speed profile, depth, sediments on the sea floor and the state of the sea [27]. This is one of the reasons that acquiring suitable training data is so hard [6]. We will describe the various properties as they relate to the sonar signal as it goes trough its normal use pattern flow. A diagram of the deployed environment of side scan sonar can be seen in Figure 2.[34].

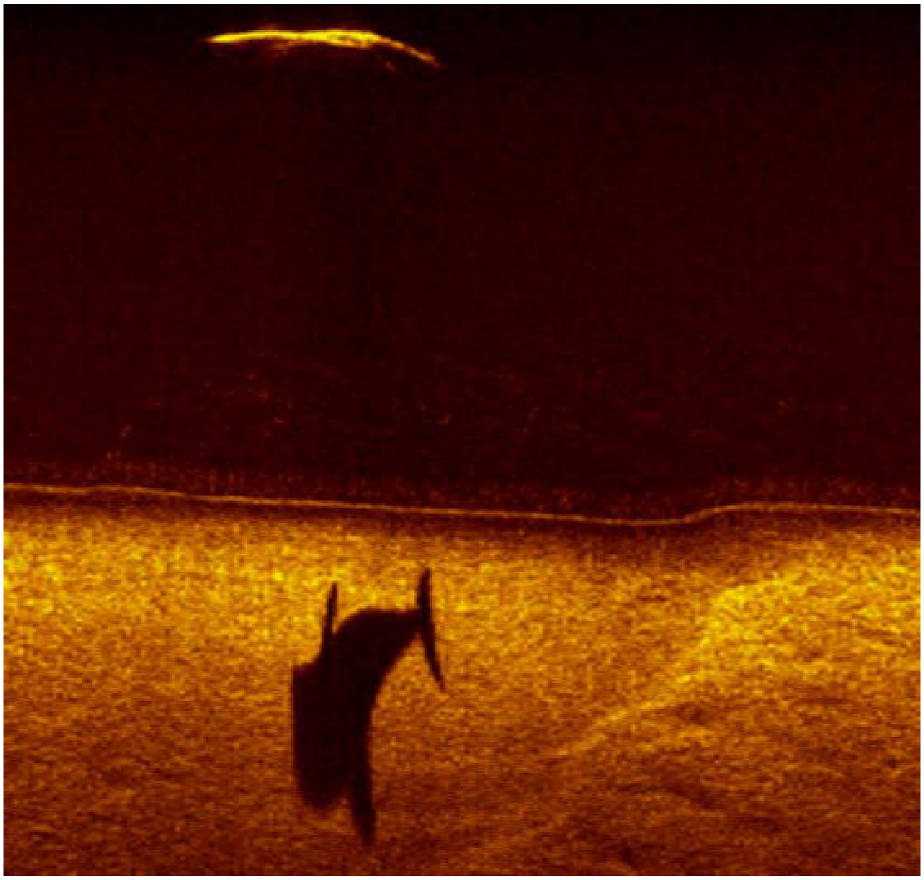

Figure 3: sonar shadow produced by a dolphin. (source: https://www.edgetech.com/underwater-technology-gallery/)

The signal first starts at the transmitter, where source level (SL) dB is measured, it is the strength of signal measured 1 meter away from the source transmitter (in decibels). The sound ray is then weakened by sonar spread loss, this is because as the sound wave gets further away from the source transmitter, it also loses amplitude and becomes weaker. In part due to this the reflected sound wave strength is much weaker than the transmitted strength. The other reason the signal is weaker is sonar sound absorption loss. To simplify a complex matter, some of the sound passing trough molecules of water gets absorbed by the molecule and converted into heat. This sound is lost and is not returned. How much gets lost in this fashion will depend on the frequency of the sound wave, with higher frequencies converting more energy to heat in this fashion than lower frequencies. It is somewhat analogous to hearing the bass (lower frequencies) of a song playing from a far away loud speaker, but not the other higher frequencies.[6]. Both of these signal modifiers (absorption and spread loss) together are called transmission or propagation loss (TL) (in decibels). Finally the sound reflection characteristics of the target detected object are known as the target strength (TS) (in decibels). [30] [34]

2.2 Automated detection from sonar images

Object detection being one of the most fundamental problems of computer vision it forms the basis question of many other tasks such as instance segmentation, image captioning, object tracking. It asks "What objects are where´´. The astounding advances in the field of deep learning have brought object detection to the forefront, making things like autonomous driving, robot vision and intelligent sensors possible [40]. An object detection model takes an input, which can be video frames for example, and then tries to predict what objects occur where. It then usually outputs pairs of where it thinks the object is (this can be a 4 dimensional bounding box vector), and the label of the objects predicted class with its predictive score (confidence), however certain approaches can also output a binary mask of the detected object. Before the advent of deep learning computer vision approaches mainly used various non learning feature based approaches such as HOG, Hough transform, SIFT and SURF. These were then used as input for traditional machine learning techniques such as SVM or k-NN algorithms. These methods are often computationally much faster, and are still used for simpler implementations. And while deep learning approaches present us with a black box system, these feature based approaches are more transparent. It should also be noted that deep learning did not render these approaches useless, as they are often used in combination with deep learning with much success [2].

The success of deep learning models in discriminative tasks has been fueled in part by the availability of big data. Deep neural networks have been successfully applied to Computer Vision tasks such as image classifcation, object detection, and image segmentation thanks to the development of convolutional neural networks (CNNs). CNNs utilize parameterized, sparsely connected kernels which preserve the spatial characteristics of images. Convolutional layers downsample the spatial resolution of images while expanding the depth of their feature maps. This allows them to create more useful representations of images across their higher and lower dimensional features than what was previously possible with networks utilizing only feed forward layers [29]. These approaches often show very promising results, but it should be noted that CNN's are usually trained on optical image data, and not images representing sonar acoustic data.

While the field of traditional optical images has gained much attention in the field of object detection. Approaches involving underwater sonar acoustic readings and object detection have been lacking behind. This is mainly due to the data being hard to obtain and usually the underwater environment being very noisy when modeled using sonar. Additional problems arise inherent to object detection. Such as the dual priorities of wanting to locate and classify the object we want to detect. This makes underwater sonar object detection a very hard problem.[17]

2.2.1 Deep learning object detection architectures

Object detection architectures are generally differentiated into two-stage or single-stage architectures based on how they approach image segmentation.

Two stage architectures include some sort of region proposal stage, where the image is segmented and areas where the object might be are proposed, these regions are then sent to a classification/regression head which classifies and locates the object inside this region (if it exists).

Single stage architectures (such as Single Shot Multibox Detector, Yolo) generally split the entire image into a grid, they then try to estimate the confidence that each of these grids contains a certain class of the object we are trying to detect. These grid cell confidences are then combined to calculate the probability of each class being present in a predicted bounding box that can span multiple grid boxes given a threshold. Two stage detector architectures (such as faster R-CNN) first try to find the number of objects in the frame (often using a region proposal network) and then classify and locate them.

The R-CNN two-stage family of object detection architectures was shown to be successful in underwater sonar human body (diver) detection [20], in autonomous underwater vehicle docking station detection [17] and it produced a better F1 score over the popular Yolov3 model in detecting humans in an autonomous underwater robot setup with optical images [37]. Specifically the Faster R-CNN implementation was shown to be able to produce state of the art results.

2.2.2 R-CNN

R-CNN stands for region based convolutional neural network and it is a two-stage object detection architecture. The faster R-CNN improves on the R-CNN architecture in terms of speed (it uses the region proposal network in place of the selective search algorithm in the Fast and normal R-CNN architectures) [28]. The region proposal network allows the model to detect regions of interest, because it needs to be trained it is more task specific, and it can produce much less detection proposals than non trained algorithms like edgeboxes and selective search while retaining a similar recall. It achieves this by using a neural network with attention that is used in a sliding windows fashion over the feature map produced by the last convolutional layer. This then finds various zones of interest using a classifier and regressor. The classifier determines the probability of a proposal having the target object, the regressor regresses the coordinates of this object. This then tells the R-CNN what regions are likely to have an object [28].

2.2.3 Detection model (Classification backbone)

Choosing a CNN model (classification head) to use with our object detection architecture depends on factors such as speed and data availability. Deep complex models take longer to train and can lead to overfitting with insufficient training data. This is because their complex multi-layered nature makes it easy to memorize a set of images by biasing its weights towards the training set, whereas a simpler model such as a shallow network does not have this capacity to this extent because of less available parameters. As mentioned before they have stronger feature extraction properties than their shallow counterparts, this makes them suitable candidates for detecting drowning victims in canals using sonar. Ge et al. [9] have used transfer learning with deep CNN models for sonar multiclass target classification (VGG-18, VGG-16 and resnet-19). They admit that there is still a problem of insufficient feature extraction ability for the classification of images with small targets and complex structures.

The receptive field size of a convolutional neural network has a large effect on the feature extraction abilities of the network. Since unlike fully connected networks, where the value of each unit activation depends on the entire input to the network, in convolutional neural networks the unit depends only on a region of the input. Because of this any input value outside of the region of input that the unit is affected by (the units receptive field) has no effect on the activation value of the unit. It follows that ensuring a sufficient size of a receptive field is very important for the predictive success of the model. One most common way to achieve this is by stacking layers to make the network architecture deeper, this increases the receptive field size linearly in theory, as an extra layer increases the receptive field size by the filter kernel size [24].

From this one might ask if learning better networks is as easy as stacking layers and whether the increased receptive field size has drawbacks. Stacking layers and therefore making networks deeper makes these models more difficult to train because of the nature of the weight updates. Since each weight receives an update proportional to the partial derivative of the error function based on the current weight, in deep networks with many weights some weights might get tiny updates that prevent the model from meaningful training. This is called the vanishing gradient problem, along with degradation of training accuracy. It is what used to make training deeper models more difficult.[11]

Residual networks were a major break trough in this domain, much like dropout (explained in the data augmentation Section) they are a way to reduce overfitting by introducing new neural network building blocks. while the theory behind the reasoning why they perform better than vanilla deep networks is somewhat disputed. They essentially added the notion of residual blocks to the deep neural network architecture. These blocks include skip connections, which allow the network to learn while not degrading the accuracy and error rate. This is achieved by adding an identity function, of the form:

$$y = F(x, Wi) + x$$

Where y is the output vector, x the input vector and the function F(x, Wi) represents the residual mapping to be learned. This allows the output of the previous layer to be added to the output of the layer after it in the residual block, alleviating the vanishing gradient problem and the degrading accuracy. This thought to be because of the new alternate shortcut path for the gradient to flow trough and the ability of higher layers to learn as good as the lower layer because of the identity function. [11]

A somewhat recent deep model based on resnet with promising feature extraction properties that is available in the pytorch and tensorflow implementations is the resnext-50 model. It is a homogeneous neural network which reduces the number of hyperparameters required and adds the notion of cardinality on top of network width and depth compared to the original resnet model. The cardinality value denotes the number of times a sequential set of transformations is performed (such as the convolutional and residual blocks described above). It can therefore increase the model complexity without adding to the width and depth of a network architecture. [36].

2.2.4 Training

As mentioned before a large dataset is crucial when using a deep CNN, as they are prone to overfit when not enough examples of a class are given. This is because complex models need many examples in order to not just memorize images, but learn useful features to help them distinguish these images. In our case we have a very large dataset, so the size is not the problem, however the class distribution is very skewed. This means that the model may not learn a useful representation of what a body would look like (its important features) in its various poses, conditions and stages of decomposition underwater because of the lack of variety in data. This might manifest as the training error decreasing while the validation set error increases. In general we want the validation error to decrease as a measure of how well our model adapts to unseen data. For this to occur our model needs to have a robust and varied representation of what a drowning victim might look like in varying underwater conditions.

Creating synthetic data can be a powerful solution to re-balance a skewed class distribution by creating more samples of the minority class [29]. An approach that proved successful would be to use a style transfer network, where sonar images are used as inputs and a style transfer network generates simulated sonar images, these simulated images are then combined with the actual training data, which improves the performance of the actual classification model over just training on the training set alone. [9]

There is also the problem of sonar readings being very prone to noise due to the nature of sound propagation in shallow muddy water. While noise in itself is not a problem when presented with a sufficiently large dataset for deep learning, the problems of applying deep learning techniques to sonar get amplified due to the lack of available training data for different underwater environments and conditions. As mentioned before, to compensate for this synthetic data is frequently used, along with approaches like transfer learning, or using pre-trained models on in-air images then fine tuning the model on sonar images [17]

[25].

An example of a pipeline that has produced very positive results includes two architectures, namely a faster R-CNN utilising a pre-trained convolutional neural network on imagenet, which is assumed to have learned the low and mid level object detection features such as edges, contours, shapes etc, and a generative model to generate simulated training data (this can also be done using a generative adversarial framework), which is used as a data augmentation technique for oversampling (to redistribute the skewed class distribution using synthetic data) [9] [20].

2.2.5 Image Segmentation

Image segmentation is the process of dividing an image into different regions based on the characteristics of pixels to identify objects or boundaries to simplify an image and more efficiently analyze it. Image segmentation can be thought of as a classification problem of pixels with labels (semantic segmentation) or separation of individual objects (instance segmentation). Semantic segmentation performs pixel-level labeling with a set of classes for all image pixels.[14]

To segment the image into regions of interest earlier implementations of the R-CNN architectures used non learning algorithms such as edgeboxes and selective search. Recent work in the domain of class independent sonar object detection has shown that these non learning methods while often fast, generate far too many region of interest proposals, and slow down detection when high recall is needed using a deep classification backbone. [20] [35]

An approach that showcases the power of image segmentation is the Mask R-CNN implementation. Mask R-CNN builds upon the Faster R-CNN architecture that was explained above. It uses the same region proposal network in the first stage, however adds a binary mask that gets predicted along with the class and bounding box predictions in the second stage. The process to calculate the mask uses a fully convolutional network to predict a m x m mask from each region of interest, this then allows for pixel-to-pixel translation between image frames which was not possible in the original faster R-CNN implementation. It has been used successfully in satellite imagery to automatically map football fields or certain types of buildings [12].

A region proposal network is a neural network that uses feature maps to generate regions where it thinks an object of a certain class is located. Using a region proposal network ensures the regions of interest can be adapted to the detection task. This is in part because of the feature network in the faster R-CNN framework which uses convolutional layers to provide the region proposal network with relevant feature maps of the input image. The generated proposals are ranked based on their "objectness score". This score estimates whether the region contains an object or is part of the background. The training aspect of this process requires data containing the target detection object, which is why simulating enough quality training samples is crucial for this approach to work, since it allows the region proposal network (and classification head that takes these region proposals as input) to get an idea of what the different poses of a drowning victim could look like in different underwater conditions. [28]

The classification head that tries to classify objects in the proposed regions should be a sufficiently deep CNN model that is able to capture small and detailed objects such as bodies with its strong feature extraction properties [9].

2.3 Simulation of data

Today the bottleneck in DL performance is often caused by the limited availability and quality of training data. No matter the potential of a particular model, the end performance will suffer if the training data cannot properly represent the distribution of data that the model will encounter [1].

Data availability is therefore often the limiting factor in training DL models, not only because of the actual process of gathering data being hard and time consuming but also because often times annotating the vast quantities of data for supervised learning can be even more time consuming and expensive. Because of these problems making the most out of the available training data has become crucial, and using data augmentation to generate synthetic data is becoming a crucial part of the deep learning pipeline to ensure model generalization, domain adaptation and adversarial robustness [1].

2.3.1 Data augmentation

While there are other ways to reduce overfitting that do not involve modifying the training data such as dropout, where a random subset of neuron activations in the network is nullified in training to prevent a small subset of neurons having large predictive power and other techniques that ensure neural network parameter regularization. These try to ensure the model learns a robust representation of features, but they do not approach the root of the problem, which is the training dataset. [29]

Data augmentation methods seek to artificially increase the training dataset size with the assumption that doing this will ensure the model learns more information from the original training dataset. There are two main categories of augmented data, one is data that has been transformed, where the original was manipulated using methods such as geometric and color transformations, random erasing and neural style transfers. These are called data warping augmentations. Oversampling augmentations on the other hand seek to create artificially generated instances of training data. These include mixing images and generating images using generative adverserial networks (GANs). It should be noted that the two are not mutually exclusive, however thought should be given to what augmentations are label preserving for supervised learning. For example rotation is a label preserving data warping technique on a sonar reading of a drowning victim, but is not a label preserving transform for training on the MNIST dataset (a 6 can get rotated into a 9) [29]

Ideally our dataset would have all classes represented somewhat equally. In our case the dataset class distribution is skewed, this means that the images of the object that we want to classify are very rare. This makes it hard for the classifier to learn what the actual object looks like in varying conditions with noise. This can be alleviated using an oversampling method, while naive simple geometric transforms can be used to generate new data, this will likely not be enough due to the varying poses that a human body can make, with varying underwater conditions. Recent approaches have proven that intelligent oversampling methods such as GANs and style transfer networks can be successful in this domain [20] [31] [9].

Data augmentation using methods such as GANs and neural style transfer can be likened to imagination or dreaming, where humans are able to imagine different scenarios based on experience. GANs and NST can imagine alterations to images such that they have a better understanding of them [29]

2.3.2 Neural style transfer (NST)

The general idea of neural style transfer is to use the representation of an image that the CNN has learned and apply the style from this representation to another image while preserving its content. The representations learned can be separated into content and style, the original paper by Gatys et al. [8] modeled the content of a photo as its feature responses from a pre-trained CNN and its style as the summary feature statistics. It was then able to iteratively recombine the content of a given image with the style of a well known artwork by trying to match desired CNN feature distributions. These involved the content information of the image and the style information of the artwork. [15]

The original style transfer algorithm has been succeeded by various improvements such as substituting the content and style loss functions and using a loss network pre-trained for image classification using a set of style images, which allows for measuring perceptual loss between images and not per pixel loss as in the original paper. These changes have produced significant improvements in terms of speed [16].

Choosing which style images to sample from is not trivial, and selecting a subset of style images that is too small can introduce further bias into the dataset. Selecting a large set of styles has computational downsides when using the method proposed by the original paper by Gatys et al.[8]. When using the faster method proposed by Johnson et al. the loss network has to be pre-trained on the style images of our choosing.[29] [16]

2.3.3 Generative adversarial networks (GANs)

GAN's simultaneously train two models, a generative model that generates samples, and a discriminative model that estimates the probability that the sample came from the training data, rather than the generative model. The generative model tries to "fool" the discriminative model by producing samples that are indistinguishable from the training data [10] [31].

The power of GANs for generating data has been shown in many domains including emotion classification, where researchers have used cycleGANs to successfully redistribute the class distribution of their training dataset [39]. GANs have also been applied to unsupervised anomaly detection, where they oversampled samples that occur with a small probability and reduced the false positive rate [22]. GANs are currently leading the way in terms of computation speed and quality of the generated data in the field of data augmentation [29].

In the realm of sonar data variables such as noise and reverberation in the shallow underwater environment play a very big role in how the final projection of the object looks like when visualized [21] [31]. To produce simulated data that is realistic and adheres to how sonar sound rays would propagate and reverberate off the object a suitable varied training dataset is needed.

2.3.4 Paired and unpaired training data considerations

In the realm of image-image translation using GANs where we are trying to capture the style of one image collection and applying those to another image collection we are often faced with the lack of paired data. Paired data refers to when we have training data of how input images map to generated images. This could be a gray-scale to color or an edge-map to photograph mappings of input images, that are then used as training data.[32] [38]

For generating paired sonar data, Sung et al. [31] used a small water tank outfitted with a Dual-frequency Identification Sonar to generate sonar images of objects and then pair them with their 3D rendered models. This approach using training data generated in a lab generalized well to test data in the sea. While paired data is useful, it is often very difficult and time consuming to obtain, Sung et al. generated 2404 image pairs [31].

A somewhat recent GAN framework that does not require paired training data and has proven very powerful in the domain of data augmentation and image to image translation is the CycleGAN framework, proposed by Zhu et al [38]. Because realistic paired training data would be very difficult and time consuming to obtain compared to using a simulator for generating input images of rendered 3D body poses, the fact that we have a very large dataset (300gb) and the very positive results of CycleGANs on rebalancing datasets with skewed class distributions [39] means that using CycleGAN to generate synthetic data is a valid approach in our domain.

2.3.5 Related acoustic data simulation approaches

There have been two promising main methods of generating synthetic data in the realm of acoustic data, and to give the reader an idea of the differences between them they will be briefly explained.

The first method (as described in [9]) with results shown in Figure 4 can be split into two parts, the first part segments the images that we want to apply the sonar style to. It splits the background and the object that we want to apply the sonar properties to (like the sonar shadow etc). The background is changed to gray, the foreground detected object then gets duplicated, the first copy is changed to white only, the second copy of the object is changed to black, and is expanded along the X or Y axis 1.2 times, this expanded black object represents the sonar shadow. The image is then fused, so the white object is superimposed onto the expanded black object with the gray background color. The second part uses a style transfer network that is trained on a set of style images and then applies what could be simplified as a “sonar style filter” over this image that is learned by the neural network. This method is somewhat crude but produced very good results when used with side scan sonar images for body detection in waters, where optical images such as those of people doing snow angels from a birds eye view were used to be simulated as sonar images of bodies in water [9].

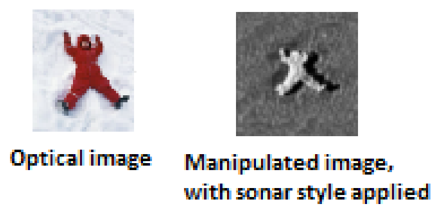

Figure 4: Sonar simulation results using first method. (source: [9])

In the second method (as described in [31]) ray tracing simulators were used to simulate how sound hitting an object would produce an acoustic shadow, and a similarly shaped real life object was placed under a sonar transmitter in lab conditions to generate a paired label image with a sonar style. Using this approach the model was able to successfully learn and map the sonar style to 3D renderings of objects as inputs. This feedback is then used in both directions as a loss function to improve the authenticity of the 3D model to sonar data generation and vice versa. This method is more computationally complex and time consuming because of the paired data but also more realistic and robust. Because of the ray tracing simulator this approach can also be used for various angles and not just top down views of objects. Essentially the simulator first emulates the imaging mechanism of the sonar sensor, which produces only the essential information such as the image highlight and shadow of the viewed scene. Then the GAN generates a more realistic image by adding degradation effects (noise, blur...) based on the object and inline with the sonar style of the training dataset. An advantage is that the same GAN trained in the opposite way was shown to help remove the degradation effects from a sonar image, and thus potentially make classification easier by allowing us to “remove” the sonar style and noise [31]. The results can be shown in Figure 5 and 6, while the resulting sonar style has some similarity to the first method (keeping in mind that the simulated sonar transmitter angle differs between them), it should be noted that the final “real” image in the second Figure is one that “fooled” the adversarial network and passed as a real image from the training sonar dataset, with the image in Figure 4 this would likely have not been the case.

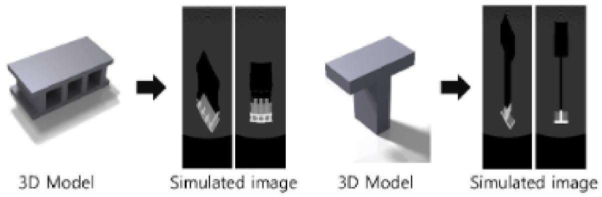

Figure 5: Ray tracing simulator using second method. (source: [31])

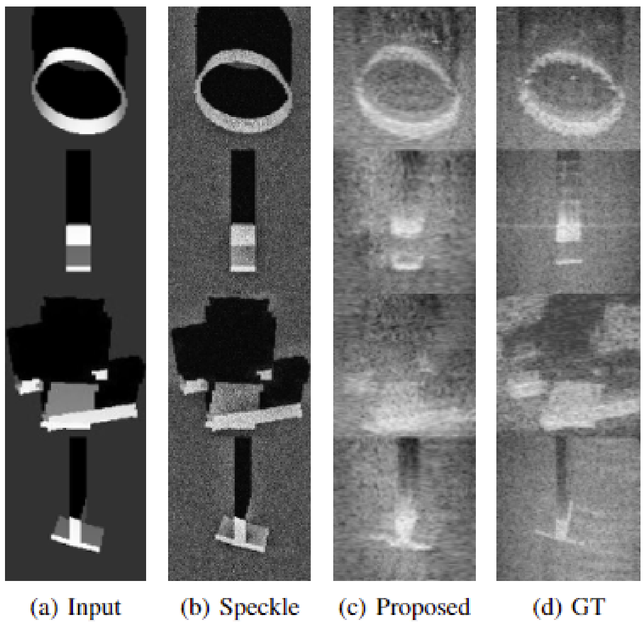

Figure 6: Sonar style applied to the simulated image. (source: [31])

2.4 Drowned body detection using sonar

While detecting human bodies using deep learning is a complicated problem, various works have been performed in the field.

Recent work focused on detecting drowning victims in sonar readings has shown promising results. Transfer learning, GANs and style transfer networks have been successfully used to simulate training data and to alleviate the problem of the small sample size of the object we want to detect [9] [20] [31]. When detecting objects in sonar readings an important factor to consider is how many frames per second the visualizing software produces. This can have an effect on model choice (accuracy vs speed/compute time trade off). However because sonar readings are slow-scrolling a real-time solution is not needed.

Huo et al [13] acknowledged the problem of lacking data and created their own dataset. The dataset contained shipwreck, drowning victim, airplane, mine and seafloor sonar images. The dataset was originally published on github with the publication of the paper but has since been removed. They acknowledge the skewed class distribution and utilize image segmentation with a geometric and color space transformation pipeline to generate synthetic data out of optical object images. They then evaluate this data with various classifiers such as using support vector machines on learned image features (BOF descriptor on SIFT features and SVM classification), a shallow CNN trained from scratch, a deep decision tree ensemble method that can alleviate the problem of small sample size and overfitting for deep neural networks (gcforrest) and two pretrained variations of the VGG16 network. They found that using synthetic data improved the overall classification accuracy for all these methods (3-4% improvements) in their object detection approach, with the fine tuned pretrained variation of the VGG16 network producing the best results compared to the same model trained only on real data.

They acknowledge that their work can be combined with recent object detection methods for multi object detection and localisation and not just classification. [13]

Ge et al. [9] focused on classifying drowning victims, aircraft, seafloor and shipwrecks from sonar images produced in the dataset by Huo et al [13]. To alleviate their class imbalance problem they firstly utilized a pre-trained CNN for detection and utilized geometric transforms to manipulate optical images to produce a grayscale image with a sonar shadow. Synthetic data was then produced by utilizing neural style transfer to apply a sonar style to these grayscale images, this is in contrast to Huo et al [13] where mainly image manipulation and segmentation was used. Utilizing transfer learning on the imagenet dataset provided the biggest accuracy benefit between variants of the vgg16, resnet18 and vgg19 models (5 to 7 %). Including the 54 generated images of drowning victims and 60 of aircraft in the training set provided a small boost in accuracy (1 to 2%). The authors suspect that utilizing a deeper network would improve results due to the insufficient feature extraction ability of shallower networks on small targets and complex structures [9].

Lee et al. [20] dealt with the problem of underwater diver detection using the traditional object detection paradigm of localising and detecting. They dealt with a lack of training data by proposing a similar approach as Ge et al. [9], where they also used deep learning for data augmentation using neural style transfer to synthetically generate data. They used StyleBankNet for neural style transfer and images produced by using 3D models with varying poses inside various UWsim environments (underwater simulator) as the input content images to StyleBankNet. StyleBankNet then used real sonar images captured from sea and water tank conditions as the style images. They split the generated synthetic training set images to grayscale and inverted single channel representation as the background of sonar readings can be darker or brighter than the detected object. The training dataset used data augmentation in the form of random flips, scale, translation and rotation variations to ensure variety of training data. They then trained the Faster R-CNN framework using a custom 4 layer deep CNN for object detection on the simulated sea and water tank conditions. They then used a human dummy to generate real sea evaluation data. They found that the model trained on simulated data with the water tank style applied achieved similar precision and recall as the model trained on real sea data. [20]

There has also been work to try to develop a robotic solution to aid in the rescue of potential drowning victims. Yaswanthkumar et al.[37] developed a raspberry pi based robot which was trained on the dataset of 800 underwater optical images, with the consequence that computation was strongly limited. This dataset was then used to train the faster R-CNN and YOLOv3 models for optical underwater human detection. When evaluated using 200 unseen images in varying pool conditions the Faster R-CNN model showed an increased F1 score (0.881) compared to its YOLOv3 baseline (0.792).[37]

3 Data

The data obtained by the police includes 300gb of sidescan sonar readings in the .jsf format. This data is split by the municipality and the name of the body of water (lake, river, canal etc). Each of these folders feature the actual sonar readings split in 51 Mb .jsf files and where objects have been investigated also a folder "Targets" that contain .jpg images of what the sonar representations of suspected drowning victims look like, and their respective EdgeTech .target files.

3.1 Side scan sonar data readings

Side scan sonar is one method used to look at the detail of the ocean floor. The sonar probe that is towed behind a boat (towfish) sends out acoustic rays from its sides at right angles to the direction the ship is moving. The tow vehicle has sensitive hydrophones (also called receivers) which receive the returning sound. The darker parts or the image represent greater echo strength. [33].

The middle of the image on Figure 7 is the path of the tow vehicle. The white line in the middle represent the sound pulse sent out from the towfish which is immediately heard by the hydrophone on the towfish. The blank white space, moving out from the black lines, is the time it takes the sound to travel through the water. The first echo from the seafloor (or sometimes from the sea surface) is the next mark. Then echos from the seafloor and objects at greater and greater distances from the tow vehicle [33]

3.2 Jsf format

The jsf format is made out of several types of messages each with a header that signifies the type of data to follow and its size. Each ping has two messages, one for the starboard side and one for the port side. In our data the most commonly

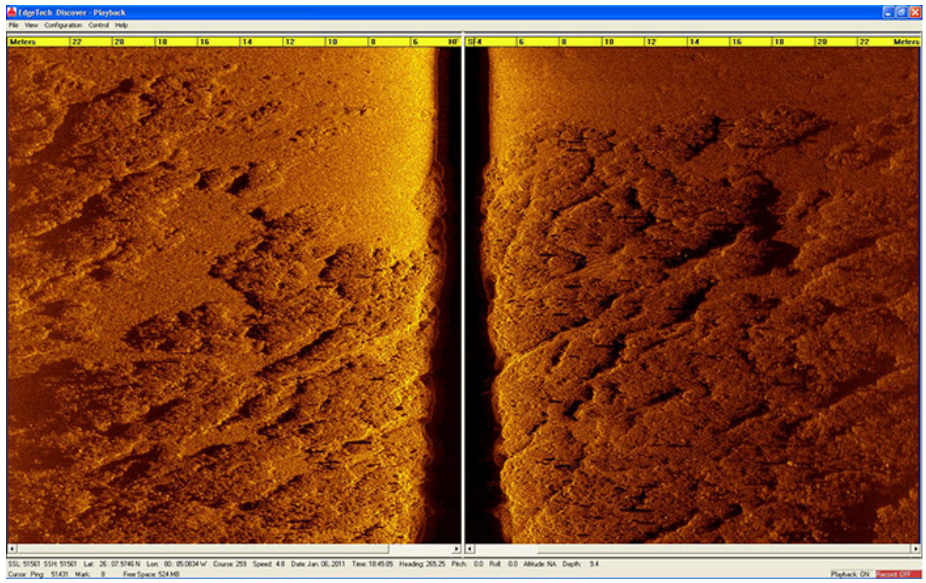

Figure 7: When visualized by EdgeTech discover, the readings by the device resemble the above image. (source: https://www.edgetech.com/underwater-technology-gallery/)

occurring and relevant messages will be discussed, as there are more than 15 message types in total, with various types of SONAR probes able to produce a set of them. Our selection includes actual sonar scan ping data messages, NMEA string messages (which are NMEA format encoded messages from either the sonar probe, the discover software or the edgetech sonar interface), Pitch roll messages (messages containing positional data related to the sonar probe such as acceleration and rate gyro in the X Y and Z dimensions, pitch and roll, rough temperature estimates), and pressure sensor messages (if a pressure sensor is installed, this contains data such as water conductivity, salinity, detailed temperature, pressure, velocity of sound and depth in meters. The messages also include system information data and navigation offset data which includes information about various offsets such as depth altitude, roll or yaw navigation offsets to name a few.

3.3 Data

3.3.1 Pre-processing

Using a well written parser for the .jsf format, some of its functionality was re-purposed and a script that stitches together multiple .jsf files chronologically was created for both starboard and port view sensors. The sonar readings are decoded using instructions and methods provided in the official documentation provided on the edgetech jsf format. The script produces clear full sonar scans of a canal survey that can span multiple seperate .jsf files. Even in cases where width greatly differs between .jsf files the script finds a max width of those files, pads all readings of a survey file collection to match that width, then scales the final stitched image to 1000px width. Effectively mimicking the behaviour of sonar visualization software. These full images have a highly variable height, which is based on the distance of the survey. Full stitched survey images often exceed 60.000 pixels and crash most standard image viewers.

A sliding window script was implemented that is able to parse these enourmous images and split them into multiple smaller training images, thus creating a dataset of images with a desired width and height. Since these sliding windows can be of resolutions that may not cover the original image fully or might produce a border over one of our samples (effectively chopping the detected body in two images) a padding parameter was added. This padding parameter ensures that no data is cut or lost. If the sliding window does not fit with the dimensions of the image (full image width is not divisible by sliding window width) then the last iterated sliding window in a row that would normally be cropped is instead produced using the image border as its end border, so that the desired dimensions are always preserved and no data is left out (example below).

This sliding window approach was applied for dataset creation with dimensions 1000x1000 pixels with a stride of 200 pixels over the entire sonar scan readings to produce sonar images. Effectively this meant that for a scan with width 1000 and height 3000 that the first image was created using a square with coordinates (x,y) 0,0 and 1000,1000. The second image was created at 0,800(1000-200) 0,1800(1000-200+1000) (200 padding overlap), the third at 0,1600(1800-200) 0,2600(1800-200+1000). The next iteration would have coordinates 0,2300 and 1000,3300, but because this is out of bounds for an image with height 3000, the last coordinates are 0,2000(image height - 1000) and 1000,3000(image height). Thus the full data was extracted while preserving 1000 x 1000 dimensions for each image.

3.3.2 Dataset preparation

For supervised learning data has to be prepared in a way where the model is able to learn important features to be able to detect novel unseen samples based on those features. The produced training images of resolution 1000x1000 were annotated using VoTT (visual object tagging tool). Objects were grouped into classes confirmed body, debris, bike and anomaly. The anomaly class represented objects that had features and shapes that resembled a body, but were not a body. The debris class was partially annotated due to the extremely large amount of debris present in the sonar data.

Ultimately all the classes except the confirmed body class were ignored due to the model producing very few predictions for this class, due to the very imbalanced nature of the dataset class distribution (see figure).

4 Synthetic data generation method

When considering a method to generate synthetic data there are various constraints that should be taken into account when choosing a valid approach.

For example an approach where an actual physical dummy is submerged in a canal might produce the most accurate synthetic data, but it requires a lot of time, money, manual work and planning.

An approach where top down optical images of a person laying on the floor are segmented and converted to gray scale with a sonar return and shadow added, then superimposed on sonar backgrounds might be easier to implement, but again, it would be very time consuming to take as many photos. The synthetic data would also likely be of very bad quality when the metric is how much it is visually similar to real sonar bodies due to the different properties of a sonar environment and an optical in air camera environment. There are of course other major considerations such as how a realistic sonar shadow would be estimated with these images, since realistic sonar shadows are usually the most important feature for detecting drowning victims.

Another approach might be to try to model the underwater scene in a 3D environment, and use light ray tracing from a light source to simulate sonar return and shadow on a generated body. The problem with this approach is that the problem of simulating a realistic underwater environment is far too large, due to the large scale of the problem there is an enormous amount of failure points.

The approach that was chosen utilizes the fact that there is a very large set of backgrounds without a positive sample, and then uses 3D modeling software to render bodies with highly varied features onto those backgrounds.

4.1 Blender

Blender is a free and open-source 3D computer graphics software tool-set that is frequently used to create animations or renderings that include 3D objects.

5 Training and hyper-parameters

6 Results

7 Conclusion

8 A dump of text to be organized

8.1 Future work

Since the models pretrained on coco and imagenet were inferior to models trained from scratch, we used models trained from scratch. In their case the width of the images is not constrained by any backbone input requirements. It is possible then to use sonar images with a higher width and thus lose less data in the preprocessing (resizing), since the body is small relative to the image anyway, resizing the image further is likely not beneficial at best.

Surveys that contain no body could easily be used just to superimpose the fake bodies onto them, and thus create much more training data with varied terrains.

Noise profiles could have been improved, with dynamic alpha blending and salt and pepper noise distributions for shadow and body noise based on background image pixel values and their distribution. This should have priority for future work, as this alone can create bodies that look indistinguishable from real data if sufficient work is done.

8.2 Results

The non pretrained model, when dealing purely with real data achieved the best recall on test set, was able to detect 16 out of the total 37 cases, with 639 false positives (TP:16, FP:639, FN:21).The same model is able to achieve 4 TP 16 FP and 33 FN at 0.8 confidence treshold.

[26] Mentions focal loss being one of the better solutions to solve the background-foreground class imbalance, but we chose to do 2 x oversampling with heavy data augmentation instead

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